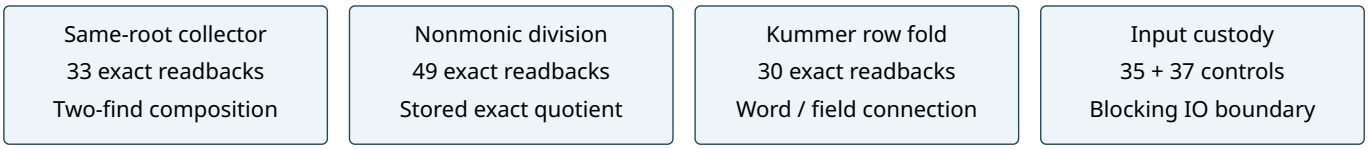
Native branch and input composition: reviewed results

Four parallel source-interface constructions now have independent reviews against published producer commit b4684a67ca626d338099525aefae817a3d276e78. All new verification used Rust, Lean or thin shell. The full frozen 1,260-target scope is preserved. This packet contains formal and fixed software controls; no scientific execution, runtime qualification, timing or new cost ratio occurred.



Same-root collector branch. The proof composes two actual find traces with opposite-sign constraints or equal-sign inverse-plus/minus-two anchors. It handles new and repeated anchors, preserves forest and global vertex/anchor equations, and derives correctness of a returned scalar. Independent replay passed 33 exact named readbacks: 6 main declarations, 12 producer controls and 15 reviewer controls. Native word/inverse composition, sign-invariant preservation, counters and allocation remain separate.

Nonmonic division. The construction derives the inverse/coefficient shortcuts, canonical remainder update, degree descent, quotient conservation, sufficient-fuel termination and exact stored word quotient. Independent replay passed 49 exact readbacks: 19 main declarations, 22 producer controls and 8 reviewer controls. Source fidelity is restricted to canonical, nonzero-leading, odd-prime degree-one no-table inputs with p less than 2^31. A malformed-input control shows that the unguarded model can return where native inversion asserts; that input is outside the theorem premises. Extended-GCD composition remains separate.

Kummer row fold. The literal word multiply/add fold is connected to the field action and widened dot-product calculation. Canonical inputs and p at most 2^32 bound the unreduced four-product dot below 2^66. Sixteen matrix-square remainder equations with canonical nonzero c imply preservation of nonzero output. Independent replay passed 30 exact readbacks: 12 main declarations, 10 producer controls and 8 reviewer controls. Counter headroom, compiler fidelity, runtime inputs and geometric action remain separate; the 105 historical records were not re-enumerated or instantiated as Lean terms here.

Native input custody. Strict raw JSON parsing and mounted-input custody checks now compose with the reviewed metadata/diff adapter. All 35 producer controls and 37 independent controls passed their expected outcomes. The review identifies unbounded directory enumeration, potentially blocking filesystem calls and a path-replacement window before File::open. Byte caps do not supply a deadline. External pin admission, later daemon mount identity and the full native supervisor lifecycle remain required interfaces.

Checks and provenance. Fresh native proof replay, exact named axiom inventories, false-theorem controls and additional reviewer controls are retained. Compiler dependency caches retain their recorded trust boundary. Reviewers disclose prior familiarity and imported-module authorship; the row-fold reviewer read execution wrappers before publication, then reviewed mathematical sources and replayed after publication. Initial proof, parser, wrong-directory and missing-source control failures remain visible and are distinguished from meaningful false controls. Earlier frozen Python evidence remains historical; new verifiers did not invoke it.

Publication and next interfaces. Inspected component PDFs, editable SVGs, structured statuses, exact source snapshots and native receipts are linked by the custody index. Compiled intermediates are omitted from publication while their original manifests and native reconstruction sources remain. Main 48cf4b7f769e29ac91430fd15acdb383846c1f66 has unchanged three_leads.rs relative to this branch. The shared journal conflict/reservation remains separate; packet-local notes record this work. Next work includes native same-root word composition and metadata bounds, extended-GCD composition, geometric/signed recovery interfaces and a bounded native supervisor before scientific continuation.